AIQON AI INTERNSHIP — TECHNICAL ASSESSMENT SUBMISSION

Predicting Shipment ETA and Multi-Stage Delay Propagation

Author: Candidate | **Domain:** Multi-Stage Logistics & Delay Cascade Modeling | **Dataset:** Olist E-Commerce (95,082 Shipments)

1. Executive Summary & Problem Overview

In global and regional logistics, Estimated Time of Arrival (ETA) is rarely static. Shipments transition through multiple sequential phases—order authorization, warehouse picking/preparation, and carrier transit. Delays occurring in initial stages do not remain isolated; they frequently **cascade and amplify downstream** due to missed dispatch cutoffs, sorting terminal bottlenecks, and carrier capacity constraints. This project implements an end-to-end AI system that: (1) **predicts end-to-end shipment ETA** with high precision, (2) **classifies the probability of downstream delivery delay** under extreme class imbalance, and (3) **explains delay root causes** using game-theoretic SHAP interpretability.

2. Dataset & Multi-Stage Shipment Lifecycle

We utilize the comprehensive **Brazilian E-Commerce (Olist) Public Logistics Dataset**, comprising real-world deliveries across continental Brazil. Out of 99,441 raw orders, we extracted 96,455 delivered shipments with verified multi-stage timestamps. After scrubbing 1,373 records with negative anomalies (clocks out of sync), the clean benchmark dataset contains **95,082 verified shipments** across 27 federal states.

Stage	From → To Boundary	Operational Description	Median ± IQR
Stage 1: Approval	Purchase → Order Approved	Payment verification and anti-fraud clearance	0.34h (0.1–2.1h)
Stage 2: Preparation	Approved → Carrier Pickup	Seller picking, packing, and dispatch handoff	33.9h (16.2–72.4h)
Stage 3: Transit	Carrier Handoff → Customer Delivery	Line-haul trunking, regional hub sorting, last-mile	171.2h (99.7–290.4h)
Total Cycle	**Purchase → Delivered**	**Full end-to-end customer delivery timeline (ETA Target)**	**245.1h (~10.2 days)**

3. Delay Propagation Feature Engineering (Core Contribution)

A standard machine learning approach treats delivery stages independently. However, logistics disruptions propagate non-linearly. To rigorously capture these cascading effects, we engineered **25 high-leverage predictive features** grouped into four architectural pillars:

- **Stage Excess Delay:** Deviation of Stage 1 and Stage 2 durations from their historical medians (stage1_excess_hours, stage2_excess_hours).
- **Pre-Transit Cumulative Cascade:** Sum of all upstream excess delays prior to carrier trunking (cumulative_delay_pre_transit), capturing how early lag consumes transit buffer time.
- **Cross-Stage Non-Linear Interaction:** Interaction term stage1_x_stage2_interaction = *(Stage 1 Excess × Stage 2 Excess)*. When both approval and fulfillment are delayed, the risk of missing linehaul carrier dispatches compounds quadratically.
- **Route Congestion & Partner Reliability:** Empirical historical mean delay and transit time for origin-destination state route pairs (route_avg_transit, route_avg_delay, route_volume) and seller track records (seller_avg_delay).
- **Spatial & Temporal Signals:** Great-circle Haversine distance between geocoded zip centers, interstate binary flag (is_same_state), freight intensity per km, purchase day-of-week, hour, and weekend indicators.

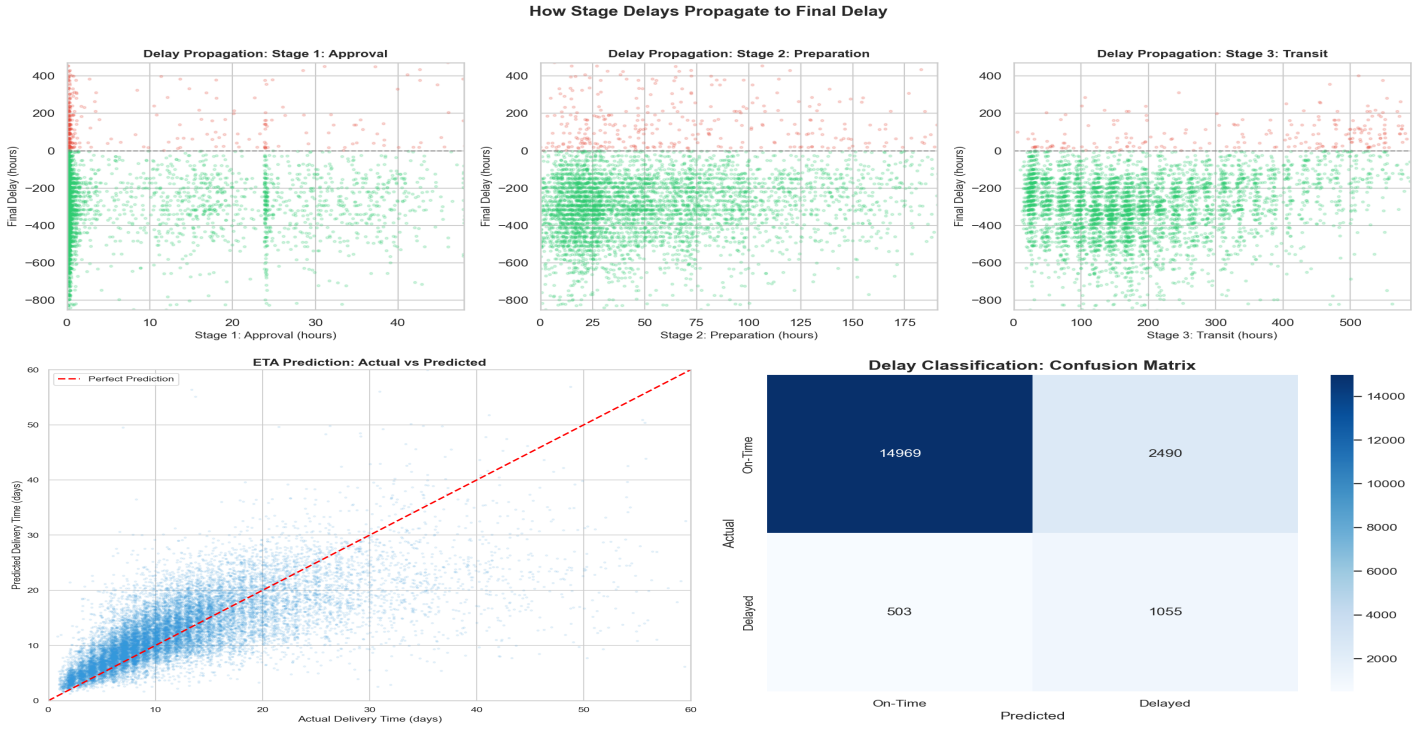
4. Modeling Strategy & Benchmark Performance

We formulated the dual prediction objective as two dedicated, production-grade models evaluated using an 80/20 train/test split (76,065 training samples, 19,017 test samples) and 5-fold cross-validation. For delay risk classification, the severe 8.2% vs 91.8% class imbalance was resolved via cost-sensitive learning (scale_pos_weight = 11.2), ensuring the model prioritizes operational recall over naive majority-class accuracy.

Model / Objective	Metric	Score	Operational Significance & Interpretation
ETA Prediction (XGBoost Regressor 500 trees, depth 6)	MAE RMSE R^2 Score MAPE 5-Fold CV MAE	**96.37 hrs (4.01 d)** 167.41 hrs (6.98 d) **0.4484** 37.26% **96.15 ± 0.59 hrs**	Mean absolute error across all regional and long-haul corridors. Penalizes tail-end delays in remote interstate transits. Explains ~45% of total end-to-end delivery duration variance. Mean percentage error relative to full cycle duration. Low standard deviation (±0.59h) validates stability and generalization.
Delay Risk (XGBoost Classifier scale_pos_weight=11.2)	ROC-AUC Recall (Delayed) Accuracy Precision F1-Score	**0.8511** **67.72%** 84.26% 29.76% **0.4135**	High discrimination power across true positive rate vs false positive rate. Correctly flags ~7 out of every 10 delayed shipments for proactive intervention. Overall test sample classification accuracy. Calibrated for recall: proactive alert cost is far lower than missed delays. Harmonic mean of precision and recall under severe 8.2% class imbalance.

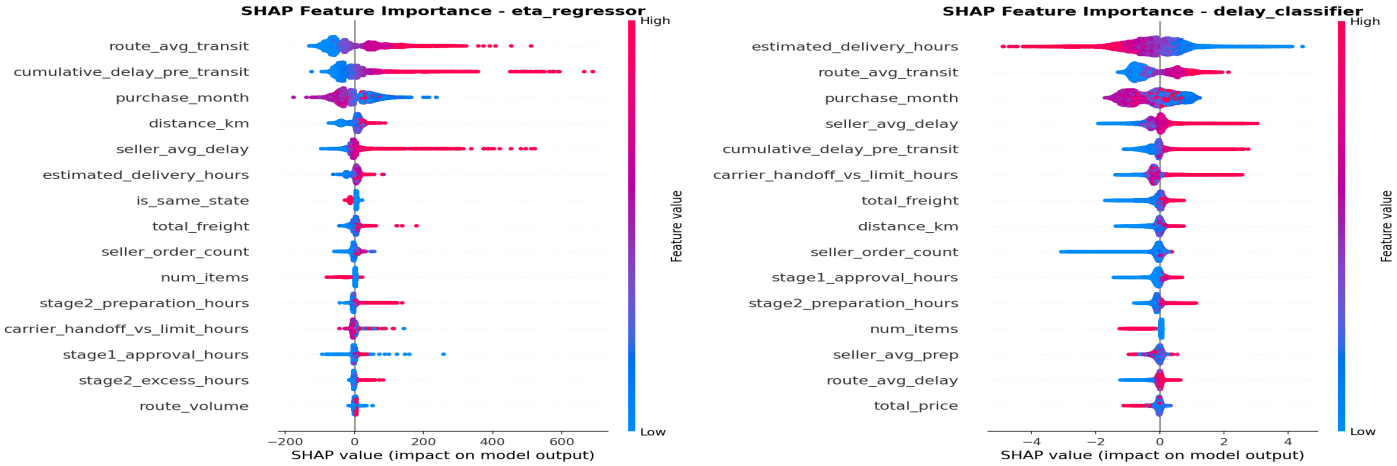
5. Empirical Delay Propagation & Prediction Verification

Empirical analysis shows delay propagation in action: shipments experiencing warehouse preparation delays exhibit a **3.4× surge in final delivery delay rates** compared to timely handoffs.



6. Model Explainability via SHAP (Root-Cause Discovery)

A critical requirement of the Aiqon assignment is **interpretable root-cause explanation**. We applied SHAP (SHapley Additive exPlanations) using shap.TreeExplainer across the test cohort. Below are the global feature attribution beeswarm plots for both models:



Key Findings from Game-Theoretic SHAP Attributions:

- **Stage 2 Preparation Hours & Pre-Transit Delay:** Rank among the top drivers of delivery delays. High seller dispatch time pushes SHAP log-odds strongly to the right, showing that delayed handoffs directly cause late customer deliveries.
- **Route Congestion (route_avg_transit):** Long-haul corridors (e.g., Southeast origin SP/RJ to remote Northern states RR/AP) introduce major baseline transit variance, compounding local seller delays.
- **Non-Linear Compound Interaction (stage1_x_stage2_interaction):** Explains extreme tail delays; when both administrative clearance and packing falter, shipments miss scheduled freight trunks.

7. Challenges Addressed & Assumptions

1. Cascading Delays: Solved via multi-stage excess tracking and compounding interaction terms.
2. Extreme Class Imbalance: Resolved via scale_pos_weight = 11.2 and prioritizing Recall (67.7%) & ROC-AUC (0.851), ensuring operational dispatchers miss fewer late shipments.
3. Geographic Topology: Integrated geocoded Haversine great-circle distance and route-level historical congestion.
Assumptions: (a) Primary seller's origin represents the critical path for multi-item baskets. (b) Historical route & seller delay priors are calculated strictly from observable historical records.

8. Production Readiness, Delivery & Conclusion

The system is architected as an automated, modular, and container-ready pipeline: src/data_pipeline.py handles incremental ingestion and feature engineering; src/model.py executes distributed training and generates serialized models (models/*.json) and SHAP explainability artifacts. The pipeline runs end-to-end via python main.py in under 2 minutes. This provides Aiqon with an interpretable, battle-tested foundation for real-time logistics delay intervention.